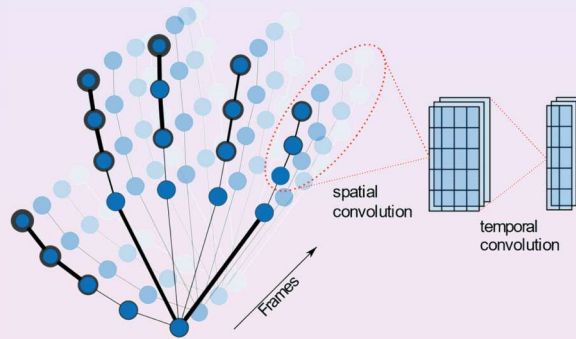


Continuous HGR for Human-Robot Collaborative Assembly

Spatio-Temporal Adaptive Graph CNN:



$$\mathbf{f}_{out} = \sum_k^{K_v} \mathbf{W}_k \mathbf{f}_{in} (\mathbf{A}_k + \mathbf{B}_k + \mathbf{C}_k) \quad (1)$$

- K_v - kernel size of spatial dimension, $\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k$ - adjacency matrixes
- $\mathbf{W}_k$ - $C_{out} \times C_{in} \times 1 \times 1$ weight vector of the 1×1 conv.
- AGCN block: spatial graph conv. followed by temporal conv.
- 7 AGCN blocks, followed by GAP, and a fully connected layer

Continuous HGR for Human-Robot Collaborative Assembly



Abstract

Motivation

Context

Previous research

Dataset:

Train: 12 videos, 54,659 RGB images: reach - 3,553, grasp - 16,414, move - 5,621, position - 10,669, release - 17,030.

Test: 3 videos, 31,190 RGB images: reach - 3,911, grasp - 10,140, move - 5,523, position - 4,044, release - 7,038

Table: Accuracies and average F1-scores.

	Accuracy [%]	F1-score [%]
LSTM	79.21	77.02
Bi-LSTM	88.99	85.40
GRU	88.65	86.01
AGCN	90.73	88.23

Continuous HGR for Human-Robot Collaborative Assembly



Abstract

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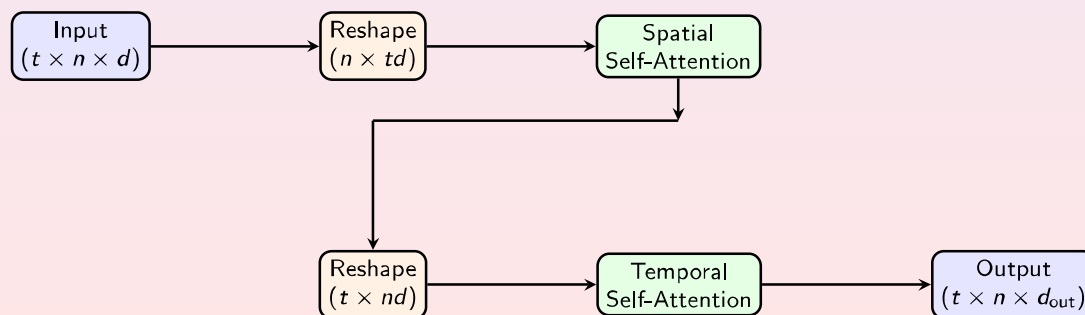
Previous research

Transformer-based Method:

- Features extracted by basic blocks of ST-AGCN
- Features extracted by basic spatio-temporal self-attention blocks.

Approach:

- Original transformer operates on sequential data, i.e., $X \in \mathbb{R}^{n \times d}$.
- For sequence of skeletal data the input is a 3-order tensor $X \in \mathbb{R}^{n \times t \times d}$,
- Data is reshaped as follows:
 - (i) reshape input to size $n \times t d$, (ii) execute spatial attention,
 - (iii) reshape $n \times t d$ to $t \times n d$, (iv) execute temporal attention.



Continuous HGR for Human-Robot Collaborative Assembly



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 - (i) reshape input to size $n \times t d$, (ii) execute spatial attention,
 - (iii) reshape $n \times t d$ to $t \times n d$, (iv) execute temporal attention.
- STSA-Net network consists of I such basic blocks, usually $I = 8$.
- Spatial or temporal position features are added to the data, which are then fed to Conv2D with 1×1 kernel and $(s, 1)$ stride:
 - In spatial encoding, the joints are encoded in turn, one after the other, and all frames in the sequence have identical encoding.
 - In temporal encoding, a given joint is encoded over time sequentially.
- Positionally encoded and embedded queries/keys are used to compute the attention weights.

STSA-Net Basic Layer: Pseudocode

Pseudocode of one STSA block

Algorithm 1: One STSA (Spatio-Temporal Self-Attention) layer.

Input: Tensor $X \in \mathbb{R}^{t \times n \times d}$

Output: Tensor $Y \in \mathbb{R}^{t' \times n \times d_{out}}$

```

1 Add spatial and temporal positional encodings to  $X$ ;
2 Spatial attention;
3    $Z_s \leftarrow \text{reshape}(X, n \times (td))$ ; // joints as sequence
4    $Q_s, K_s, V_s \leftarrow \text{Linear}(Z_s)$ ;
5    $A_s \leftarrow \text{softmax}(Q_s K_s^\top / \sqrt{d_k})$ ;
6    $H_s \leftarrow \text{Linear}(A_s V_s)$ ;
7    $H_s \leftarrow \text{reshape}(H_s, t \times n \times d')$ ;
8 Temporal attention;
9   (Optional) downsample along time:  $H_s \leftarrow \text{Conv1D}(H_s, \text{stride} = s)$ ;
10   $Z_t \leftarrow \text{reshape}(H_s, t' \times (nd'))$ ;
11   $Q_t, K_t, V_t \leftarrow \text{Linear}(Z_t)$ ;
12   $A_t \leftarrow \text{softmax}(Q_t K_t^\top / \sqrt{d_k})$ ;
13   $H_t \leftarrow \text{Linear}(A_t V_t)$ ;
14   $Y \leftarrow \text{reshape}(H_t, t' \times n \times d_{out})$ ;

```

Abstract

Motivation

Context

Previous research

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ST-Transformers for Skeleton-Based Action Rec.



Model		Year	Key Idea / Contribution
ST-TR (SpatialTemporal Transformer)		2021	Plizzari et al. Treat joints as tokens. Spatial attention (across joints) followed by temporal attention (across frames).
MS-TR (Multi-Scale Transformer)	Trans-	2021	Liu et al. Multi-scale joint partitioning + dual transformer branches capturing short- and long-term dependencies.
PoseFormer		2021	Zheng et al. Transformer encoder on skeleton tokens; designed originally for pose estimation but adapted for action recognition.
DSTA (Dynamic Spatio-Temporal Attention)		2022	Shi et al. Dynamic spatial attention for joints + temporal attention; transformer replaces graph convolution.
STAR-Transformer		2023	Ahn et al. Cross-attention between skeleton tokens and (optional) RGB tokens; strong HAR performance.
STMT (SpatialTemporal Mesh Transformer)		2023	Zhu et al. Applies transformer on mesh/skeleton representations; disentangles spatial and temporal attention.
STSA-Net		2023	Kwolek. Reshape → spatial self-attention → reshape → temporal self-attention. Lightweight and suitable for small datasets.

Abstract

Motivation

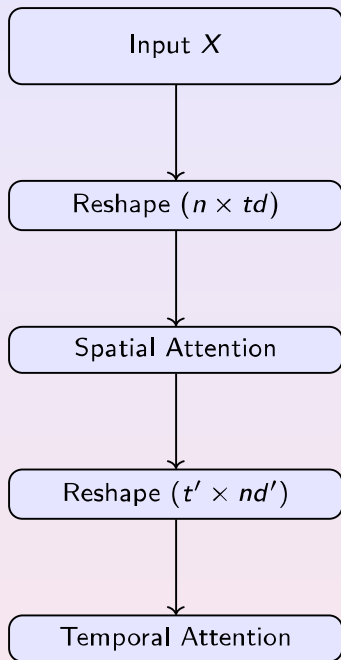
Context

Previous research

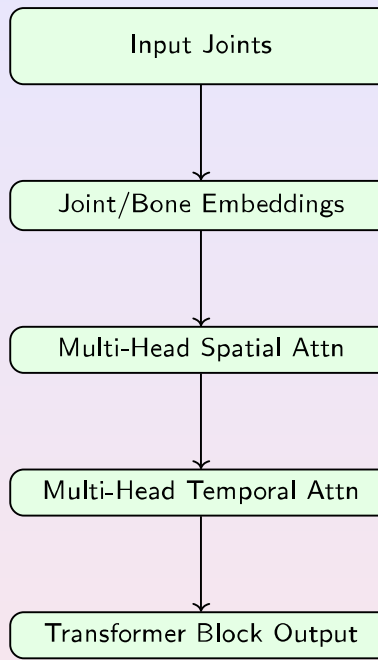
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STSA-Net vs. ST-TR: Visual Comparison



STSA-Net (2023)



ST-TR (2021)

STSA-Net (ICCVW 2023)

Lightweight reshape-based spatial→temporal attention
 No joint graph prior; operates on flattened joint features
 Designed for continuous gestures
 Simple linear layers, optional temporal down-sample
 Optimized for efficiency

ST-TR (ICPR 2021)

Heavy transformer with multi-head spatial and temporal attention
 Implicit graph reasoning via joint/bone embeddings
 Designed for large-scale action recognition (NTU-RGB+D)
 Deep transformer encoder blocks with strong capacity
 Optimized for representational power

Abstract

Motivation

Context

Previous research